

Question Number	Answer	Additional Guidance	Mark
2(a)	there will be no difference between the length of leaves treated with insecticide or water (after 56 days) (1)		(1) exp

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	26 / 26.1 / 26.13 (%)	Allow 21 / 20.7 / 20.71	(1) grad

Question Number	Answer	Additional Guidance	Mark																																																			
2(b)(ii)	- suitable table format with data (1) - correct column headings with units (1) - means correctly calculated (1)	Length of leaf /cm <table><tr><th></th><th>Insecticide treated</th><th>Water treated</th></tr><tr><td></td><td>12.6</td><td>9.1</td></tr><tr><td></td><td>13.1</td><td>9.4</td></tr><tr><td></td><td>13.9</td><td>10.3</td></tr><tr><td></td><td>14.0</td><td>10.1</td></tr><tr><td></td><td>11.1</td><td>10.4</td></tr><tr><td></td><td>11.6</td><td>10.9</td></tr><tr><td></td><td>12.4</td><td>9.0</td></tr><tr><td></td><td>12.9</td><td>9.7</td></tr><tr><td></td><td>12.5</td><td>9.9</td></tr><tr><td></td><td>13.4</td><td>10.8</td></tr><tr><td></td><td>13.7</td><td>9.3</td></tr><tr><td></td><td>12.0</td><td>9.8</td></tr><tr><td></td><td>11.9</td><td>9.2</td></tr><tr><td></td><td>12.6</td><td>9.7</td></tr><tr><td></td><td>11.8</td><td>9.4</td></tr><tr><td>mean</td><td>12.6</td><td>9.8</td></tr></table>		Insecticide treated	Water treated		12.6	9.1		13.1	9.4		13.9	10.3		14.0	10.1		11.1	10.4		11.6	10.9		12.4	9.0		12.9	9.7		12.5	9.9		13.4	10.8		13.7	9.3		12.0	9.8		11.9	9.2		12.6	9.7		11.8	9.4	mean	12.6	9.8	(3) exp
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2c	• bar graph with linear scale and axes labelled with unit (1) • means plotted correctly (1) • range bars plotted correctly (1)	ALLOW ECF from 2b axes mean length (cm) and insecticide /water 12.6 9.8 14-11.1 and 10.9 -9.0	(3) exp

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	• numerator calculation or value shown (1) • denominator calculation or value shown (1) • $t = 10.3 / 10.31$ (1)	= 'insert calc' / 2.8 OR 'insert calc' / 2.83 = 'insert calc' / 0.272 OR 'insert calc' / 0.273 OR 10.4 / 10.37 Correct answer with no working gains full marks	(3) exp

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	An answer that includes two of the following points: • critical value is 2.048 and the calculated value (10.31) is more than the critical value (1) • therefore reject the null hypothesis there is a difference between the length of the leaf of insecticide treated and water treated(thistle plants) (1) • comment on variability of data (1)	IGNORE other calculated values as long as they are greater than 2.048 range bars do not overlap IGNORE reliability / accuracy / reliability / repeatability	(2) exp

Question Number	Answer	Additional Guidance	Mark
2(e)	An answer that includes two of the following points: - the leaves should have been measured before the treatment (because the rate of growth may be different) (1) - the environmental conditions need to be controlled/measured (because they will affect the rate of growth) (1) - the leaves should have been measured at different {time interval(s) / growth season} (to compare the rate of growth) (1) - use of different concentrations of insecticides (1)	IGNORE different fields ALLOW measured from the same part of the plant eg at a specified height or 5th leaf down ALLOW a named condition controlled/measured ALLOW in a greenhouse/eq Allow a longer period of time	(2) exp

(Total for question 2 = 15 marks)